Math 103 Class 3

1. A <u>frequency distribution</u> is the organizing of raw data in table form, using classes and frequencies.

2. **Rules for Constructing a Frequency Distribution**

   a.) There should be between 5 and 20 classes.
   b.) The class width should be an odd number.
   This insures the midpoint will have the same place value as the data. Midpoints are used for graphing and for mean and standard deviation.
   c.) The classes must be mutually exclusive.
   (nonoverlapping limits)
   d.) The classes must be continuous.
   No gaps.
   A class with a zero frequency can be omitted at either end of the distribution.
   e.) The classes must be exhaustive.
   All data must be accommodated.
   f.) The classes must be equal in width.

3. **Procedure for Constructing a Grouped Frequency Distribution**

   a.) Find the highest and lowest value.
   b.) Find the range.
   c.) Select the number of classes desired.
   d.) Find the width by dividing the range by the number of classes and rounding up.
   e.) Select a starting point (usually the lowest value); add the width to get the lower limits.
   f.) Find the upper class limits.
   g.) Find the boundaries.
   h.) Tally the data, find the frequencies, and find the cumulative frequency.

4. **Reasons for Constructing a Frequency Distribution**

   a.) To organize the data in a meaningful, intelligible way.
   b.) To enable the reader to determine the nature or shape of the distribution.
   c.) To facilitate computational procedures for measures of average and spread.
   d.) To enable the researcher to draw charts and graphs for the presentation of data.
   e.) To enable the reader to make comparisons among different data sets.

5. **Example for Class Practice: Grouped Frequency Distribution**

10	08	06	14
22	13	17	19
11	09	18	14
13	12	15	15
05	11	16	11

1. Find the highest and lowest values

2. Find the range

3. Select the number of classes desired

4. Find the class width. Divide the range by the number of classes.

5. Select a starting point for the lowest class limit. For convenience, this value is chosen to be 5, the smallest data value. The lower class limits will be:

6. The upper class limits will be:

7. Find the class boundaries by subtracting 0.5 from each lower class limit and adding 0.5 to the upper class limit.

8. Tally the data. Fill in the frequency column. Determine the Relative Frequency by dividing the frequency by the total

9. Fill in the cumulative frequency. Determine the cumulative relative frequency.